

Review Notes for CoEvoP&R

Organization. Each manuscript section is reviewed in its own part. Comments under one part apply only to that manuscript section unless a cross-section consistency issue is stated explicitly.

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Part I

Abstract Review

1 Overall Assessment

The abstract has a strong overall structure: it moves from the problem, to the limitations of prior work, to the proposed method, its safety and evaluation mechanisms, quantitative results, and the final deployable artifact. The central novelty—searching the differentiable placement objective itself—is clear and compelling. The length and overall progression are also appropriate. There are no serious grammatical problems, but the abstract would benefit from several stylistic clarifications and a few substantive checks before it is finalized.

2 Wording and Clarity Comments

1. “Fixed expert terms” would read more naturally as “fixed, expert-designed terms.”
2. “Scheduled routed promotions condition later proposals” is accurate but difficult to understand without first reading the Method section. A plainer formulation, such as “results from scheduled routing evaluations inform subsequent proposals,” would be clearer.
3. “Pass a restricted objective interface” is awkward. Candidates pass validation checks imposed by the restricted interface; they do not literally pass the interface itself.
4. “Gives cheap timing proxies selection influence” is grammatically possible but dense. It would be clearer to say that the audit “allows inexpensive timing proxies to influence selection.”
5. The evaluation list is not parallel: “ChiPBench Nangate45, ICCAD 2015 Superblue, and ASAP7 cross-node transfer” mixes benchmarks, design families, technology nodes, and an experiment type. A clearer construction is: “We evaluate CoEvoP&R on the ChiPBench Nangate45 and ICCAD 2015 Superblue benchmarks and study cross-node transfer to ASAP7.” This wording should be checked against the exact experimental setup.
6. “Plugs directly into DREAMPlace” may imply that no integration layer is required, whereas the Method describes restricted lowering into an `ObjectiveSpec`. “Executes within DREAMPlace through the restricted objective interface” may be more precise.
7. The abstract contains several specialized phrases in quick succession, including “scheduled routed promotions,” “term-compatibility rules,” and “perturbation-based proxy audit.” The proxy audit is worth retaining if it is a principal contribution, but these mechanisms should be described as intuitively as possible in the abstract.

3 Substantive Concerns

1. **The reported evidence does not fully match the opening motivation.** The abstract motivates the work using routed wirelength, routing congestion, and post-route timing, but reports improvements only in HPWL, density overflow, and placement-RC timing. A reviewer may immediately ask whether routing feedback improves routed outcomes. If a strong post-GRT result is available, at least one routed-quality result should be reported in the abstract.

2. **The scope of the numerical claims is ambiguous.** It is unclear whether the reported 16.9%, 36.7%, 0.70 ns, and 912 ns improvements are averages over designs, seeds, or both. In the current sentence, “on average” appears to modify only HPWL and density overflow. The scope of every number should be made explicit.
3. **The direction of the TNS result should be explicit.** If TNS is stored as a negative quantity, “improving TNS by 912 ns” can be difficult to interpret. Depending on the metric convention, “reducing TNS magnitude by 912 ns” or another directionally explicit description may be better.
4. **The opening correlation claim may be too broad.** The phrase “correlate poorly with routed wirelength, routing congestion, and timing” appears to make one general claim about all three metric categories. Unless the cited evidence demonstrates weak correlation for every listed metric, a more defensible formulation would be “do not consistently predict” or “can correlate poorly with.”
5. **The benchmark terminology must be exact.** The roles of ChiPBench, Nangate45, Superblue, and ASAP7 should be described consistently as benchmark suites, design families, technology platforms, or transfer settings, as appropriate.
6. **The central novelty should remain dominant.** The strongest distinction is that prior work tunes parameters, adds fixed terms, or uses learned surrogates, whereas CoEvoP&R searches the complete differentiable objective program. This should remain the clearest and most memorable claim in the abstract.

The benchmark wording, numerical scope, and routed-quality claim should be verified against the Experiments and Results section before the abstract is finalized. These points have not yet been validated because the present review stops at the end of the Method section.

4 Proposed Revision

The following revision improves the wording while preserving the technical claims currently made in the manuscript. The numerical and benchmark language should still be checked against the experiments, and a routed-quality result should be added if available.

Analytical placers typically optimize half-perimeter wirelength (HPWL) and density, although recent post-route benchmarks show that these proxies can correlate poorly with routed wirelength, routing congestion, and timing. Existing routing-aware placers address this mismatch using fixed, expert-designed terms or learned black-box surrogates, while leaving the form of the differentiable objective unchanged. CoEvoP&R instead searches over complete replacement objectives. A large language model (LLM) proposes readable symbolic objective programs, DREAMPlace evaluates their behavior within the optimization loop, and results from scheduled routing evaluations inform subsequent proposals. Each candidate passes a restricted objective interface, term-compatibility checks, autograd validation, and a cost-scaled placement, timing, and routing evaluation cascade. A perturbation-based audit allows inexpensive timing proxies to influence selection only when they provide information beyond HPWL and agree with downstream timing. We evaluate CoEvoP&R on the ChiPBench Nangate45 and ICCAD 2015 Superblue benchmarks and study cross-node transfer to ASAP7. On the

primary ChiPBench benchmark, CoEvoP&R reduces HPWL by 16.9% and density overflow by 36.7% on average relative to native DREAMPlace, while improving OpenTimer placement-RC WNS by 0.70 ns and TNS by 912 ns. Each search produces a compact symbolic objective that can be inspected, edited, and executed within DREAMPlace through the restricted objective interface.

Part II

Introduction Review

5 Review Scope

The perspectives below are lenses used to evaluate the Introduction; they are not all topics that must be added to the Introduction itself. The Introduction should primarily establish the problem, summarize the relevant limitations of prior work, define the research gap, introduce CoEvoP&R, state its contributions, and summarize its most important evidence. Detailed safety, reproducibility, archive, and evaluation procedures belong in the Method and Experiments sections.

This review intentionally ignores the keyword list and reference formatting.

6 Overall Assessment

The Introduction has a sound narrative arc: proxy mismatch, limitations of existing routing-aware approaches, objective-program search, contributions, and results. The main idea is understandable without first reading the Method section. It does not require restructuring from scratch, but several claims should be made more precise and the reported evidence should better match the routing-centered motivation.

7 EDA and Technical-Accuracy Perspective

1. Calling both HPWL and density “proxies” is debatable. HPWL is a wirelength proxy, whereas density is primarily a feasibility or overlap control objective. A more precise opening would be:

Modern analytical placers make global placement tractable by minimizing a smooth wirelength surrogate together with a density objective.

2. The statement that ChiPBench shows poor correlation with routed wirelength, congestion, WNS, TNS, and DRC count should be checked carefully against what that work demonstrates for each metric. If the evidence is not uniform across all metrics, “does not consistently predict” or “can correlate poorly with” is safer than a categorical claim.
3. “The analytical placer optimizes one target while the completed flow depends on another” is memorable but overly singular. The completed flow depends on several downstream objectives. A more precise alternative is:

Consequently, the placer optimizes objectives that do not fully represent the quality metrics of the completed physical-design flow.

4. The two-part classification of routing-aware placement is useful, but it should not imply that every method fits perfectly into either a hand-designed symbolic term or a learned surrogate. The wording should present these as two important lines of work rather than an exhaustive taxonomy.

5. “Require expert redesign for each technology or benchmark family” is too absolute. Hand-designed objectives can transfer, although they may require expert retuning or redesign. The latter formulation is easier to defend.
6. “Black-box models that engineers cannot inspect or edit” is also too strong. Engineers can technically inspect architectures and modify learned models. The intended contrast is interpretability and direct editability:

Learned surrogates provide downstream awareness but are generally less interpretable and directly editable than compact symbolic objectives.

8 LLM and Program-Search Perspective

1. The central novelty should be stated as precisely as possible: CoEvoP&R searches the functional form of the differentiable objective while keeping the placement optimizer fixed and evaluating each candidate within the optimizer’s gradient path. This is stronger and clearer than merely saying that an LLM is used for placement.
2. “Complete symbolic replacement objectives” may sound like unrestricted program synthesis. The Method later shows that programs use a restricted grammar, an allowed observable library, and required wirelength and density anchors. The Introduction should establish this boundary early, for example:

CoEvoP&R searches compositions and nonlinear interactions among a restricted set of differentiable placement observables.

3. “This embedding makes optimizer compatibility part of the search space” is imprecise. Compatibility is a validity requirement or evaluation constraint, not necessarily a dimension of the search space. A better formulation is:

Embedding candidates in the optimization loop makes optimizer compatibility an explicit validity requirement.

4. The Introduction should explain briefly why an LLM is suitable for this task: the search space contains readable program structures and nonlinear compositions, while automatic EDA measurements provide concrete revision feedback.
5. The term “co-evolving” should be explained. At present, the objective programs evolve while routing measurements provide feedback; routing itself does not appear to form a second evolving population. One sentence should clarify what the name CoEvoP&R is intended to denote.

9 Conference-Reviewer and Adversarial Perspective

1. The distinction from prior LLM-based placement work must remain explicit. Prior work searches parameters, optimizer components, or placement actions; CoEvoP&R searches the differentiable objective program itself.

2. The Introduction should avoid winning the novelty argument by overstating weaknesses of prior work. The phrases most likely to attract reviewer objections are “cannot inspect or edit,” “require expert redesign,” and any implication that all previous differentiable objectives are entirely fixed or unsearchable.
3. The claim that the method improves “post-GRT quality” needs direct support in the Introduction. The reported summary currently includes HPWL, density overflow, and placement-RC timing, but no routed metric. Since routing feedback appears in both the motivation and the paper title, at least one post-GRT wirelength, congestion, overflow, or timing result should be reported if available.
4. The Introduction should distinguish placement-RC timing from post-route timing. A reviewer may object if the former is used as evidence for a broad post-route timing claim.
5. The contribution list must contain claims that are individually verifiable later in the paper. In particular, “discovers readable formulas that improve placement and post-GRT quality” should correspond to explicit downstream experiments.

10 Experimental-Methodology Perspective

1. The final numerical sentence should state what is averaged: designs, seeds, or both. It should also clarify whether “on average” applies to the WNS and TNS improvements as well as HPWL and density overflow.
2. The direction of the 912 ns TNS result should be explicit. If TNS is represented as a negative quantity, consider reporting a reduction in TNS magnitude or otherwise stating the sign convention.
3. The baseline wording should be precise. “Relative to native DREAMPlace” is clearer than “reduces DREAMPlace HPWL,” because HPWL is the metric and DREAMPlace is the baseline tool or objective.
4. The result summary should reflect the principal scientific claim. If routing-aware objective evolution is the key contribution, a routed outcome should accompany the placement metrics.
5. The Introduction need not describe the full evaluation cascade, but it should avoid language that implies final-panel information influences the search if the Method explicitly keeps search and final evidence separate.

11 General-Reader and Language Perspective

1. The second paragraph contains three individual prior-work result claims for LaMPlace, GOALPlace, and RoutePlacer. These establish that prior methods are effective, but they interrupt the Introduction’s argument and may fit better in Related Work. A compact alternative is:

Learned approaches such as LaMPlace, GOALPlace, and RoutePlacer have demonstrated substantial timing and routability improvements.

2. “Later congestion and density controls build on the same idea” is vague. It should either identify the relevant development more concretely or be removed.
3. The third paragraph introduces too many mechanisms in rapid succession: program evolution, gradient-path execution, static admission, autograd checks, runtime guards, scheduled routing promotion, and proxy auditing. Preserve the objective-search distinction and defer lower-level mechanisms to the Method section.
4. “Scheduled promotions return routed evidence to later proposals” and “gives cheap timing signals selection influence” are difficult for a general reader. A clearer summary is:

Placement, timing, and scheduled routing measurements are returned as feedback for subsequent proposals.

5. The first and fourth contributions overlap because both emphasize readable symbolic objectives. The first should focus on the search formulation; the fourth should focus on the inspectable, editable artifact returned by the search.

12 Recommended Revision Priorities

The Introduction is conceptually solid. Revision should prioritize the following changes:

1. Make the characterization of prior work more measured and defensible.
2. Define the restricted objective-program search space accurately.
3. State the novelty as search over the objective’s functional form with the optimizer held fixed.
4. Explain the meaning of “co-evolving.”
5. Add routed experimental evidence that supports the motivation and title, if such evidence is available.
6. Clarify the averaging scope and direction of all reported results.
7. Reduce implementation detail in the introductory method summary.

Part III

Related Work Review

13 Review Scope

This part reviews the substance and organization of the Related Work section. It focuses on technical accuracy, fairness to prior work, coverage of the most relevant research directions, the clarity of the novelty comparison, and readability. Citation and reference formatting are intentionally excluded.

14 Overall Assessment

The Related Work section is concise and organized around three useful themes: routing-aware placement, automated placement search, and executable LLM program evolution. It includes several of the closest comparisons, especially RoutePlacer, EvoPlace, VeoPlace, Eureka, and LLM-SR. Each paragraph also tries to identify the artifact modified by prior work and contrast it with CoEvoP&R’s objective-program search.

The main weakness is that several technically different methods are compressed into categories that are too broad. Reviewers familiar with these papers may challenge the resulting descriptions. The section should make its comparison axes more precise and devote more attention to the closest work instead of listing many broad LLM-for-EDA examples.

15 Technical-Accuracy Perspective

1. **The historical attribution to RePlAce should be corrected.** “RePlAce established the electrostatic density formulation” is inaccurate or at least misleading. The electrostatics-based placement formulation is generally associated with ePlace; RePlAce subsequently advanced this analytical-placement framework. A safer formulation is:

ePlace introduced the electrostatics-based placement formulation, which RePlAce and DREAMPlace subsequently advanced through improved optimization and GPU acceleration.

If ePlace is not discussed elsewhere, it should likely be added here.

2. **RUDY should be described as a routing-demand estimator.** RUDY is not inherently a differentiable objective term. It estimates routing demand from placement geometry, and later placement methods can incorporate RUDY-derived quantities into congestion-aware objectives. A more precise formulation is:

Hand-designed routing estimators such as RUDY approximate routing demand from placement geometry and can be incorporated into congestion-aware placement objectives.

3. **LaMPlace, GOALPlace, and RoutePlacer should not be presented as though they use equivalent mechanisms.** Their roles should be mapped explicitly:

- LaMPlace learns pixel-level masks for cross-stage metrics in macro placement.
- GOALPlace learns post-route-informed density targets using an empirical-Bayes approach.
- RoutePlacer trains a graph neural network as a differentiable approximation of routability and integrates it into gradient-based placement.

The current phrase “through masks, graph neural networks, and learned routability signals” does not clearly associate each mechanism with its method.

4. **The comparison with LaMPlace should acknowledge task scope.** LaMPlace concerns macro placement and generates masks for sequential placement decisions, whereas CoEvoP&R operates inside a mixed-size analytical placement objective. Treating both simply as learned downstream surrogates hides this important distinction.
5. **The first heading does not match all of its contents.** “Routability-Aware Objectives” begins with RePLace and DREAMPlace, which are foundational analytical placers rather than specifically routability-aware objective methods. A clearer heading is “Analytical and Routability-Aware Placement.”
6. **The second heading is too narrow.** “Foundation-Model Placement Search” also includes reinforcement learning and Bayesian optimization, neither of which is necessarily foundation-model search. A better heading is “Automated and Foundation-Model Placement Search.”

16 Novelty-Positioning Perspective

1. **EvoPlace deserves a sharper comparison.** It appears to be one of the closest works in terms of LLM-guided evolution around analytical placement. The essential distinction should be stated directly:

EvoPlace searches optimizer implementation components while retaining the placement objective; CoEvoP&R fixes the optimizer and searches the objective program.

2. **Eureka is also methodologically close.** Its generated reward functions execute during a learning loop, so the statement “These systems evaluate generated programs after execution” is too broad. The distinction is not simply execution after the fact versus execution inside an optimizer. A more defensible comparison is:

Unlike prior program-evolution systems, CoEvoP&R must admit generated objective expressions into an existing differentiable EDA optimization loop, requiring finite values, valid gradients, and stable behavior throughout placement.

3. The novelty should be organized around the *artifact modified*: parameters, macro-placement actions, optimizer components, learned surrogates, or the differentiable objective form. This is the cleanest basis for distinguishing CoEvoP&R from the closest methods.
4. The section should avoid implying that using an LLM or executing a generated program is itself the central novelty. The stronger claim is the search over restricted, differentiable objective programs evaluated inside the fixed analytical optimizer and informed by downstream routing evidence.

17 Consistency with the Method Section

1. “CoEvoP&R places generated code inside DREAMPlace’s autograd path” conflicts slightly with the Method statement that raw model code is never imported or executed. The Related Work section should refer to “lowered objective expressions” or “validated symbolic programs.”
2. “Before routed validation” suggests that every valid objective is routed. According to the Method, only scheduled Tier-C promotions receive routed evaluation. A more accurate formulation is:

before placement evaluation and possible promotion to routed validation.

3. “Scheduled routed promotions condition later objective proposals” is unnecessarily method-specific and difficult to parse. “Uses scheduled routing results as feedback for subsequent objective proposals” is clearer.
4. Terminology should remain consistent across sections, particularly “routing-aware” versus “routability-aware,” “post-GRT” versus “post-route,” and “generated code” versus “lowered objective program.”

18 Coverage and Balance Perspective

1. The routing-aware paragraph moves directly from RUDY to three learned methods. This may appear selective to a domain reviewer. It should briefly acknowledge other important classical directions, such as cell inflation, congestion-aware density control, global-routing feedback, pin-density or pin-accessibility terms, and differentiable congestion estimation.
2. These classical approaches do not all require detailed discussion. A single synthesis sentence can demonstrate awareness of the broader field and prevent the two-category framing from appearing constructed only to motivate the proposed method.
3. The broad LLM-for-EDA examples—ChipNeMo, RTLCoder, MasterRTL, and VeriLoC—are less directly relevant than EvoPlace, Eureka, LLM-SR, or RoutePlacer. If space is limited, shorten the general survey and use the recovered space for deeper comparison with the closest approaches.
4. ChiPBench is an evaluation benchmark rather than a placement-search method. Its role in validating whether placement gains survive downstream flow is relevant, but this role should be separated clearly from the list of systems that search parameters, optimizer components, or actions.

19 General-Reader and Language Perspective

1. Several paragraphs read as lists of papers followed by one sentence about CoEvoP&R. More synthesis would help the reader understand the design dimensions and tradeoffs instead of merely recognizing method names.
2. Ending every paragraph with a short CoEvoP&R contrast is useful but repetitive. The paper could synthesize the comparisons in a concluding paragraph or compact table, allowing the individual paragraphs to describe prior work more neutrally.

3. “Readable symbolic objective forms” is somewhat subjective. “Compact symbolic objective forms” or “restricted symbolic objective programs” is more measurable and consistent with the Method.
4. The mechanism descriptions should map explicitly to individual works instead of relying on the order of method names and technical phrases.

20 Recommended Organization

A clearer three-part organization would be:

1. **Analytical and routability-aware placement.** Cover foundational analytical objectives, hand-designed routing estimators, congestion-aware techniques, and learned downstream guidance.
2. **Automated search for placement.** Organize methods by the artifact searched: parameters, placement actions, optimizer code, or objective form.
3. **Executable program evolution.** Discuss reward-function, heuristic, symbolic-equation, and code evolution, followed by the differentiability and EDA-validation requirements specific to CoEvoP&R.

If space permits, a compact comparison table could make the principal distinctions easier to verify:

Method family	Artifact modified	Feedback	Output
AutoDMP and HyperPlace	Hyperparameters	Placement metrics	Parameter configuration
VePlace	Macro actions	Placement metrics	Placement
EvoPlace	Optimizer components	HPWL	Optimizer code
RoutePlacer	Learned routability model	Routing labels	Neural surrogate
CoEvoP&R	Differentiable objective form	Placement, timing, and routing	Symbolic objective

21 Recommended Revision Priorities

1. Correct the historical attribution of electrostatics-based placement and consider adding ePlace.
2. Describe RUDY as a routing-demand estimator rather than inherently as an objective term.
3. Separate the mechanisms and scopes of LaMPlace, GOALPlace, and RoutePlacer.
4. Deepen the comparisons with EvoPlace and Eureka.
5. Replace “generated code” with terminology consistent with the restricted lowering interface.
6. Broaden the classical routing-aware coverage with one concise synthesis sentence.
7. Reduce broad LLM-for-EDA examples if space is needed for closer work.

Part IV

Method Review

22 Review Scope

This part reviews the Method section, Algorithm 1, Figure 1, and Figure 2. It separates fundamental logical inconsistencies from lower-priority notation, reproducibility, and presentation issues. The review does not use or assess the Experiments and Results section.

23 Overall Assessment

The Method has a strong conceptual architecture. Its most promising elements are complete objective replacement, a restricted symbolic language, optimizer-embedded validation, cost-aware evaluation, negative memory, and an auditable archive. The separation between the LLM, the controller, and the EDA tools is also valuable.

The principal problem is not a lack of technical depth. Several sophisticated mechanisms are described in parallel without all of their mathematical and algorithmic connections being closed. These inconsistencies should be resolved before sentence-level proofreading because they affect the core claims of the paper.

24 Critical Internal Inconsistencies

24.1 Figure 1 Contradicts Complete Objective Replacement

The Method explicitly states that Equation 2 defines the complete minimized objective and is not a residual around a hidden native loss. Figure 1 instead depicts

$$f_{\theta}(\mathbf{x}; d) = f_{\text{base}}(\mathbf{x}; d) + \sum_j \alpha_j h_j(T(\mathbf{x}; d)),$$

which is a residual added to a base objective. This directly contradicts one of the paper’s central novelty claims. Figure 1 should instead visualize the restricted complete objective, for example,

$$f_{\theta}(\mathbf{x}; d) = \kappa_{\text{wl}} F_{\theta}(\tilde{\mathcal{T}}(\mathbf{x}; d)).$$

This is the most urgent figure correction.

24.2 The Timing-Proxy Weight Does Not Affect Selection

The Method defines a timing-proxy weight γ_q in the proxy-admission equation, but γ_q never appears in the Tier-A selection score, final score, parent policy, or archive-update equation. Therefore, the mathematical method does not currently support the claim that timing proxies affect selection.

The timing contribution must enter an explicitly defined score, eligibility rule, ranking, or parent policy. For example, if consistent with the intended implementation, a term of the following form could be incorporated into a selection score:

$$\sum_q \gamma_q w_q g_q(f).$$

The algorithm introduces a second ambiguity by running Tier B “only if proxy-admitted.” Tier-B evaluation appears necessary to obtain the proxy values being audited. The paper must distinguish admission of a global proxy signal from admission of an individual candidate to Tier B; otherwise, the logic appears circular.

24.3 Routed Feedback Does Not Formally Affect Parent Selection

The text states that each routed promotion can change parent selection, retrieval, or failure context. However, the parent probability depends only on the Tier-A score, distance from a fixed mechanism, and mechanism frequency. No Tier-C or routed quantity appears in the parent policy.

Routed results can clearly enter later prompts through feedback reconstruction, but their effect on parent selection is not defined. The paper should choose one of two interpretations:

1. State that routed evidence affects retrieval and proposal context but not parent selection; or
2. Add a routed-evidence term, eligibility update, or archive update that explicitly changes the parent pool or parent weights.

Because routing feedback is central to the title, this connection must be unambiguous.

24.4 The Search and Final Evidence Split Is Ambiguous

The text says final-panel metrics never enter prompts or archive fitness, but the returned objective is selected using S_{final} over the routed subset of the final archive. It is unclear whether “final” means the archive at generation G , search-panel Tier-C evidence, or a held-out final evaluation panel.

If held-out final-panel metrics select f^* , the panel is no longer a purely held-out reporting panel. A clearer formulation would define separate panels,

$$\mathcal{D}_S = \text{search panel}, \quad \mathcal{D}_F = \text{held-out final panel},$$

select using search-panel routed evidence,

$$f^* = \arg \max_{f \in \mathcal{A}_G^{C,S}} S_C^S(f),$$

and evaluate the already selected f^* on $\mathcal{D}_F$ only for final reporting.

24.5 The Complexity Tie-Break Claim Is Mathematically Incorrect

The final score subtracts $\eta_C c(f)$ from the routed-rank score and then says that complexity matters only when ranks tie. This is not generally true: the complexity term can reverse a non-tied rank ordering unless η_C is bounded appropriately.

If complexity is intended only as a tie-breaker, use a lexicographic rule:

1. Maximize the routed-rank score.
2. Among tied candidates, minimize complexity.

Alternatively, provide a bound proving that the complexity penalty cannot overcome any nonzero rank difference.

25 Problem Formulation

1. Clarify whether each term scale κ_j is specific to a design, seed, objective, or complete search run.
2. Specify which placement and objective call determine the “first-call” scale and how zero or extremely small values are handled.
3. Clarify whether κ_{wl} is design-specific and how it is computed.
4. Define exactly what constitutes a wirelength or density “anchor.” It is unclear whether an anchor must appear additively or may occur only inside a nonlinear interaction.
5. Explain how the panel-specific metric scales ν_j are chosen and demonstrate that their construction does not use held-out final-panel information.
6. State whether budgets count candidates, complete panels, individual designs, or design-seed runs. The mathematical text calls them placement, timing, and routing “calls,” while the algorithm increments once per candidate.
7. The notation F_θ may suggest trained parameters. If θ represents an LLM-proposed program structure and constants rather than learned parameters, say so explicitly.
8. “Keeping the optimizer fixed” should specify which elements remain unchanged: initialization, learning-rate schedule, density-weight schedule, preconditioning, legalization, detailed placement, and term implementations. A reviewer will want to know whether objective replacement indirectly changes any optimizer-specific schedules.
9. Clarify whether WNS and TNS are treated as signed negative quantities so that $o_j = +1$ is directionally correct.

26 Archive-Conditioned Proposal

The proposal mechanism is promising but requires additional definitions for reproducibility.

1. Define d_{base} , the distance metric, and the set of “fixed mechanisms” against which distance is measured.
2. Define a mechanism signature and explain how signatures are extracted from the lowered program.
3. Define the bounds that permit a candidate to enter the near-miss pool.
4. Give the cardinalities and selection criteria for Top, Diverse, NearMiss, and RelevantFail.
5. Explain how relevant failures are matched to a selected parent’s terms or mechanisms.
6. State the number and coefficient values of deterministic siblings and how their coefficient ranges are selected.
7. Give the maximum number of provider retries and identify the sampling temperature, randomization settings, and decoding limits.

8. “Prevents repeated sampling” is too strong. The mechanism-count denominator discourages repeated sampling but does not prevent it.
9. Distinguish static/schema repair retries from autograd rejection, placement failure, and completed metric regression. The phrase “malformed or rejected responses” could imply that every failure type triggers an immediate provider retry, while Figure 2 suggests that some failures enter negative memory instead.
10. Clarify whether the claimed K independent samples remain independent after provider repair retries and whether deterministic provider behavior is controlled through explicit seeds when available.

27 Safe Objective Lowering

The separation between raw responses and lowered expressions is a major strength. However, the searched language and numerical gates require more formal specification.

27.1 Define the Allowed Term Library

The Method names weighted-average and log-sum-exp wirelength, electric and bell-shaped density, soft-RUDY, long-net pressure, pin density, and pin-count-weighted pressure, but does not provide their mathematical definitions. Because the objective program is the central searched artifact, the paper should provide a table or appendix containing, for every allowed term:

- Mathematical expression
- Required inputs
- Smoothing constants
- Normalization
- Gradient behavior
- Computational cost
- Availability conditions

Without these definitions, the reader cannot reproduce or fully assess the objective search space.

27.2 Clarify the Restricted Language

1. “The action space is finite” may be too strong if numeric constants can take arbitrary values. “Restricted action space” or “bounded program structure” is safer unless the literal set is explicitly finite.
2. Define “safe division,” including its epsilon or clamping rule.
3. State whether AST depth and primitive limits apply per expression or to the complete program, including all local assignments.
4. Define allowed coefficient ranges and explain how they were selected.

5. Explain how baseline clones are detected: structural equality, canonicalized AST equality, or numerical equivalence.
6. Define exactly how “one wirelength family” and “one density family” are counted when terms occur multiple times or inside nonlinear expressions.

27.3 Strengthen the Dynamic Safety Definition

The condition

$$0 < \|\nabla_{\mathbf{x}} f(\mathbf{x}_0)\|_2 < \infty$$

may reject a valid objective if the initialization happens to be stationary. Conversely, a very large but finite gradient passes. Consider explicit lower and upper numerical thresholds, possibly normalized against the native objective.

The paper should also clarify whether the dynamic gate is applied independently for every design and seed. Because the first-call test establishes only local executability, the notation $\mathcal{F}_{\text{safe}}$ may overstate the guarantee. “Locally executable candidates” would be more precise unless a stronger property is established.

Runtime guards should specify whether they test only finiteness or also detect finite but exploding objectives and gradients.

28 Cost-Scaled Evaluation

The three-tier architecture is sensible, but this subsection contains several conceptually different mechanisms. It would be clearer if divided into:

1. Evaluation cascade
2. Tier-A feasibility and scoring
3. Timing-proxy audit
4. Routed promotion and final selection

28.1 Evaluation Cascade

1. “OpenROAD/ChiPBench” makes ChiPBench sound like a routing engine. Describe the relationship precisely, such as an OpenROAD flow configured or evaluated through ChiPBench, if that matches the implementation.
2. Clearly distinguish FLUTE/Elmore timing from OpenTimer placement-RC timing and state which one is used during search, candidate selection, and final reporting.
3. State whether placement-RC timing is evaluated before or after legalization and which parasitics, libraries, clocks, and constraints are used.
4. Clarify whether every promoted Tier-C candidate is evaluated on the same designs and seeds so that downstream comparisons are genuinely matched.

28.2 Tier-A Gate and Score

1. Define “structural failure” and “severe regression,” including all thresholds.
2. Explain whether worst-design gains first aggregate placement seeds or whether the minimum is taken over every design-seed pair.
3. Define the rank averaged by $\bar{r}_A$: the ranked quantity, direction, treatment of ties, and aggregation across metrics and design-seed cells.
4. Define the complexity function $c(f)$ and its scale.
5. “The strongest HPWL/overflow member” is ambiguous because a fixed portfolio can have a Pareto frontier rather than one strongest member.
6. Formalize the additional fixed-portfolio and Pareto gate. It currently appears only in prose and is not represented in χ_A or the algorithm.
7. Explain why HPWL and density enter both as direct gains and through the rank term, and whether this intentionally gives them additional weight.

28.3 Timing-Proxy Audit

The perturbation audit needs a reproducible definition of the perturbations and statistics. Specify:

- What is perturbed
- How perturbations are sampled
- Number of samples
- Whether correlations are pooled across designs or computed per design
- Minimum sample count
- Treatment of ties and missing timing values
- Exact computation of $a_C(q)$
- Meaning of “preserve rank direction”
- Justification for the 0.70 and 0.95 thresholds

Low correlation with HPWL establishes only non-redundancy with HPWL; it does not establish that a timing proxy is informative rather than noisy. The routed agreement test is therefore essential and should be specified rigorously.

The phrase “held-out routed promotions” should also identify what those promotions are held out from and confirm that they are distinct from the final reporting panel.

28.4 Routed Promotion and Final Ranking

1. Define how high- S_A candidates and occupants of uncovered mechanism cells are balanced within a promotion batch.
2. Define “matched downstream evidence,” particularly when some post-GRT metrics are unavailable.
3. Define $\bar{r}_C$: metrics included, orientations, design and seed aggregation, treatment of ties, and treatment of missing values.
4. Ensure that candidates are not rewarded for missing difficult metrics merely because the score averages only “reported” metrics.
5. Consider whether a feasibility or worst-case gate is required before final rank averaging so that a catastrophic regression in one routed metric cannot be hidden by improvements in others.

29 Archive and Closed-Loop Feedback

The detailed archive record is a strength and supports the auditability claim. The MAP-Elites and feedback mechanisms nevertheless require further specification.

1. Define all bin boundaries for objective size, HPWL, density, and timing.
2. Explain whether mechanism labels are determined from declared term usage, parsed AST structure, or observed behavior.
3. Define the treatment of hybrid mechanisms and missing timing.
4. Give deterministic tie-breaking rules for candidates with equal scores in one cell.
5. Use island-specific notation in the archive-update equation or explain how the global archive and island archives relate.
6. Define migration frequency, number of migrants, and whether copied elites retain their original island identity and lineage.
7. Avoid overloading the notation m_j , $m(f)$, and $\mathbf{m}_g$ for an evaluator metric, mechanism category, and feedback scorecard.
8. Explain how “repair lessons” are generated. State whether they are deterministic templates derived from failure codes or model-generated summaries.
9. Clarify which archive fields a routed result can change. This is necessary to support the claim that routing evidence changes the closed-loop search rather than merely being stored for later display.
10. “Provider snapshot” may be difficult to guarantee for closed model APIs. Specify exactly which provider and model-version metadata can be archived.

30 Algorithm 1

Algorithm 1 is useful, but it does not yet fully match the prose and equations.

1. The number of islands I is used but is not listed as an input.
2. The initial feedback packet $\mathbf{m}_0$ is not defined.
3. Dynamic finite-value and autograd admission is not shown.
4. Provider retries and their budget are omitted.
5. The lowered candidate set $\mathcal{S}_g$ defined earlier is never used; the algorithm instead introduces raw responses $\mathcal{Y}_g$ and lowers them again.
6. The Tier-B “proxy-admitted” condition must be reconciled with the proxy-audit procedure.
7. The timing weight γ_q never affects insertion, scoring, parent sampling, or archive updates.
8. Budget increments should state whether a multi-design panel counts as one evaluation or multiple design-seed calls.
9. Once B_A is exhausted, only the innermost calibration loop breaks; the outer candidate and generation loops continue. The intended termination behavior should be shown.
10. Migration scheduling should explicitly use τ_{mig} .
11. The algorithm should define behavior when no candidate receives valid Tier-C evidence and $\mathcal{A}_G^C$ is empty.
12. Held-out final-panel evaluation is absent. Add it after selection if the panel is intended only for reporting.
13. The algorithm should show the fixed-portfolio/Pareto elite gate if that gate affects archive insertion or routed eligibility.

Algorithm 1 should be revised only after the timing score, routed-feedback path, and evidence split are settled; otherwise, it will need to be rewritten multiple times.

31 Figure 1 Review

31.1 Strengths

- The proposal, tools, and memory stages make the high-level loop intuitive.
- The closed feedback path is visually prominent.
- Negative memory and the MAP-Elites archive are represented clearly.

31.2 Concerns

1. Replace the residual-objective formula, which contradicts complete objective replacement.
2. Tier C should be labeled “scheduled routed promotions” rather than “selected finalist.” Final selection occurs only after routed evidence is collected.
3. The Tier A $\rightarrow$ Tier B $\rightarrow$ Tier C flow implies that every Tier-C candidate passes through Tier B. This does not clearly match the prose, where timing is conditional and Tier-C promotion is based primarily on Tier-A elites and mechanism coverage.
4. Mutation labels such as “recombine” and “simplify” should appear in the text if they are actual controller modes. Otherwise, use only the modes implemented and described.
5. The five-dimensional MAP-Elites archive is drawn as a two-dimensional grid. Label it as a schematic projection.
6. Several labels are likely to be too small at final two-column print size. Verify the minimum font size in the compiled paper.
7. The caption should explain the green, gray, and red paths and identify which failures are retried versus stored as negative memory.
8. “BRAIN” may sound anthropomorphic. “Proposal,” “Evaluation,” and “Archive and Feedback” would be more technical stage names.
9. The figure should show explicitly whether routed evidence changes parent scoring, archive eligibility, or only the next prompt context.

32 Figure 2 Review

32.1 Strengths

- The figure clearly separates the response schema from the executable record.
- The rejected-candidate paths are useful.
- The example helps make the restricted objective-program concept concrete.

32.2 Concerns

1. The caption implies that schema, AST, term, and autograd checks all lower a response into `ObjectiveSpec`. The diagram and Method instead place autograd validation after lowering. Distinguish static lowering from dynamic admission.
2. The finite-value and autograd gate occurs on the first `DREAMPlace` objective call, not entirely before `DREAMPlace` evaluation. The figure should show the gate inside or at the boundary of the first `DREAMPlace` call.
3. The example term `soft_rudy_pnorm` should match the term names and categories used in the prose.
4. The prompt tuple should use the same field ordering and notation as the corresponding equation unless a reason for the difference is stated.

5. Schema validation appears both in “Schema-bound JSON” and in the later grammar/schema check. Remove the apparent duplication or explain the two distinct schema checks.
6. Static retry failures and dynamic negative-memory failures should have visually distinct destinations.
7. Text is likely too small when the figure is printed at one-column width. Verify readability in the compiled manuscript rather than only in the source PDF.
8. Clarify whether only a lowered `ObjectiveSpec` enters the optimizer, consistent with the statement that raw model code is never executed.

33 Structural and Presentation Recommendations

1. Divide Cost-Scaled Evaluation into explicit subsections for the cascade, Tier-A score, proxy audit, and routed promotion.
2. Divide Archive and Closed-Loop Feedback into separate archive and feedback subsections.
3. Avoid placing two section labels on one subsection; give the proxy audit and search procedure their own visible headings.
4. Add a compact notation table. The Method introduces many sets, scores, scales, counts, gates, and archive descriptors before all are defined.
5. Add a term-library table containing the objective observables and their formulas.
6. Include one concrete candidate objective example in the text and trace it through lowering, Tier-A evaluation, possible routing promotion, and archive insertion.
7. Reduce forward references such as “score defined below” where a short reordering would make the procedure easier to follow.

34 Recommended Revision Order

1. Correct Figure 1’s objective formula.
2. Formalize the search-panel and final-panel split.
3. Connect γ_q to an actual selection rule.
4. Decide whether routed evidence affects parent selection or only prompt context.
5. Correct the final complexity tie-break.
6. Define the objective-term library and normalization scales.
7. Align Algorithm 1 with the corrected logic.
8. Revise Figure 1 and Figure 2 to match the algorithm precisely.
9. Only then perform sentence-level condensation and proofreading.

The section’s underlying idea remains strong. Once the missing mathematical connections are made explicit, the Method will become substantially more convincing, reproducible, and easier to review.